

UNIT 1: Software Development Life Cycle

Introduction

SDLC defines stages in software development.

Detailed Explanation

Phases

Planning, design, implementation, testing, and maintenance.

Applications

Used in project development.

Summary

This unit introduces SDLC.

UNIT 2: Requirements Engineering

Introduction

Focuses on gathering and analyzing requirements.

Detailed Explanation

Requirement Types

Functional and non-functional.

Applications

Used in system design.

Summary

This unit explains requirements.

UNIT 3: Design Concepts

Introduction

Design defines system architecture.

Detailed Explanation

Principles

Modularity and abstraction.

Applications

Used in software development.

Summary

This unit introduces design concepts.

UNIT 4: Testing Strategies

Introduction

Testing ensures software quality.

Detailed Explanation

Types

Unit, integration, and system testing.

Applications

Used in quality assurance.

Summary

This unit explains testing.

UNIT 5: Maintenance and Project Management

Introduction

Maintenance ensures software longevity.

Detailed Explanation

Types of Maintenance

Corrective, adaptive, and perfective.

Applications

Used in project management.

Summary

This unit explains maintenance.

20CS4T04 – Computer Networks

UNIT 1: Network Fundamentals

Introduction

Computer networks connect devices for communication.

Detailed Explanation

Types of Networks

LAN, WAN, and MAN.

Applications

Used in communication systems.

Summary

This unit introduces networking basics.

UNIT 2: Data Link Layer

Introduction

Ensures reliable data transfer.

Detailed Explanation

Functions

Error detection and correction.

Applications

Used in communication protocols.

Summary

This unit explains data link layer.

UNIT 3: Network Layer

Introduction

Handles routing and addressing.

Detailed Explanation

Protocols

IP protocol.

Applications

Used in internet communication.

Summary

This unit explains network layer.

UNIT 4: Transport Layer

Introduction

Ensures end-to-end communication.

Detailed Explanation

Protocols

TCP and UDP.

Applications

Used in reliable communication.

Summary

This unit explains transport layer.

UNIT 5: Application Layer

Introduction

Provides user-level services.

Detailed Explanation

Protocols

HTTP, FTP, SMTP.

Applications

Used in web and email services.

Summary

This unit explains application layer.